Arnav Singhal

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Data Scientist specializing in deep learning for medical imaging, biomedical NLP, and ML systems deployment. Reduced reconstruction error by 34% in graduate thesis research on Fourier-augmented neural architectures, and built end-to-end pipelines spanning dynamic pricing, biomedical question-answering, and cardiovascular imaging. Skilled in Python, PyTorch, and cloud-based ML workflows (GCP), with a track record of translating research into deployed, containerized applications. Recently completed M.Sc. in Germany and now seeking Data Scientist roles in India.

Education

M.Sc. Data and Computer Science | Universität Heidelberg | Heidelberg, Germany *April 2023 - March 2026*

Major: Machine Learning and Deep Learning | Minor: Economics

Thesis: Evaluating Fourier Layers in DeepPET Architecture — A Comparative Study for Forward Modelling of the 2D Helmholtz Equation

- Designed an FFT-augmented U-Net (DeepPET) integrating spectral convolution layers with hybrid fusion and skip connections for 2D wave-field reconstruction.Best-performing architecture variant reduced mean relative error by 34% and validation loss by 62% versus the baseline model, validated via multimode evaluation and physics-aware metrics in TensorBoard.

B.Tech. Computer Science and Engineering | Galgotias University | Greater Noida, India *2016 – 2020*

Experience

Data Analyst | Toluna Corporate | Gurugram, India *June 2021 – September 2022*

- Optimized data processing workflows on the Quantum survey analytics platform, improving processing efficiency and data integrity for client deliverables.
- Led a team of 5 junior analysts, sustaining a 100% client satisfaction (CSAT) score across delivery cycles.
- Delivered insights to Procter & Gamble with 0 client-reported errors, earning consistent top ratings for accuracy, quality, and timeliness.

Jr. Data Analyst | E2E Research Services Pvt. Ltd | New Delhi, India *January 2020 – May 2021*

- Used SPSS and advanced Microsoft Excel to track, merge, and analyze survey datasets for client reporting.

Core Competencies

- **ML Focus Areas:** Medical Imaging, Scientific ML, Biomedical NLP, Explainable AI, Predictive Modelling, ML Deployment
- **Machine Learning & Deep Learning:** Supervised & Unsupervised ML, Neural Networks, NLP, Predictive Analytics
- **Data Engineering & Processing:** Python, PyTorch, SQL
- **Cloud & MLOps:** Google Cloud Platform (GCP), Weights & Biases, Docker, Git
- **Model Deployment & Tools:** Streamlit, SimpleITK, Quantum (survey analytics platform)
- **ML Lifecycle Management:** Model training, experiment tracking, model monitoring
- **Libraries & Tools:** NumPy, Pandas, Matplotlib, SPSS, Excel
- **Research & LLMs:** Prompt Engineering, Transformers, BERT Models, Biomedical Text Mining, RAG
- **Business & Governance:** Stakeholder Alignment, Data-Driven Decision-Making
- **Languages:** English (Fluent), German (A2)

Key Projects

Simulated Dynamic Pricing Engine | *Python, ML, Streamlit, Docker | GitHub*

- Built a real-time pricing engine that predicts ride fares from contextual signals: demand, distance, weather, vehicle type, and area.
- Engineered a cost structure for 10k+ samples [briefly: base fare, per-km rate, demand multiplier, etc.] to generate realistic training targets.
- Trained and compared [models tried, e.g. Linear Regression, Gradient Boosting] and selected a Random Forest regressor (RMSE 2.08, MAE 1.67, R^2 0.49) on a [80/20] held-out split.
- Deployed an interactive Streamlit dashboard in Docker so users get instant price estimates for any input scenario, reproducibly.

Explainable Emotion Detection Using B-Cos Networks | *Deep Learning, Computer Vision, PyTorch | FER2013 | GitHub | Heidelberg University*

- Built and compared a standard CNN and a B-cos CNN, which builds interpretability into the convolution layers, for 7-class facial emotion recognition on FER2013 (28,709 training and 7,178 test 48×48 grayscale images).
- Trained both models with identical settings (SGD with momentum, cross-entropy loss, batch size 64, learning-rate scheduling, horizontal-flip augmentation and normalization), so any difference came from the architecture alone.
- Evaluated with training and validation curves, per-class confusion matrices, and gradient-based visualizations of the facial regions each model relied on.
- Found that the B-cos CNN matched the baseline's accuracy, with small gains on happiness and anger. Its maps focused on the eyes and mouth, while the standard CNN's heatmaps were diffuse. This shows interpretability without an accuracy cost, which matters in healthcare and human-computer interaction.

PubMedQA: Biomedical Literature Search System | *NLP, Biomedical AI, Streamlit, HuggingFace Transformers | GitHub | Heidelberg University*

- Built an end-to-end question-answering system over 9,999 PubMed abstracts (2013-2023) that lets users ask questions in natural language and get a direct answer instead of a list of articles.
- Indexed each article's title and abstract, keyed by PMID, as dense vectors using txtai with PubMedBERT embeddings, so retrieval matches on meaning rather than keywords.
- Designed a two-stage retrieve-then-extract pipeline. Semantic search finds the k closest articles, and a DistilBERT (SQuAD-distilled) QA model extracts the answer from their combined text.
- Deployed it as an interactive Streamlit app with a question box and a search button.
- Evaluated with Jaccard similarity against reference answers, using a threshold to score correctness [test set size 3 questions]. Analysis of the failures pointed to weak context handling and irrelevant retrieval on broad queries such as "AI's role in healthcare".

Coronary Arteries Reconstruction from CTA Dataset | *Medical Imaging | Heidelberg University*

- Processed CTA volumes (.mhd) with SimpleITK and extracted artery centerlines using a CNN-based orientation tracker.
- Generated 3D vessel reconstructions (M×N×4) from the tracked centerlines.
- Achieved 2.71 mm mean centerline deviation, with 100% of points within 5 mm of ground truth.
- Explored stenosis simulation to model narrowed vessels for cardiovascular analysis.

Certifications

- Hands-On Data Science | Certificate
- Student Innovation Lab – International Student Support Prototype | Heidelberg University — completed Entrepreneurship I & II, Leadership & Team Dynamics, and Experience Design Thinking coursework; designed user journeys and prototypes using experience-driven product development.
- Seminar: Evidentiality-Guided Generation for Knowledge-Intensive NLP Tasks — Retrieval Augmented Generation, Summer 2024

Impact & Recognition

- Employee of the Month — Toluna Corporate (May 2022)
- Above and Beyond Award — Toluna Corporate (May 2022)
- Consistent 5-Star Performance — Delivered exceptional service to Procter & Gamble with 0 client-reported errors and top ratings for accuracy, quality, and timeliness